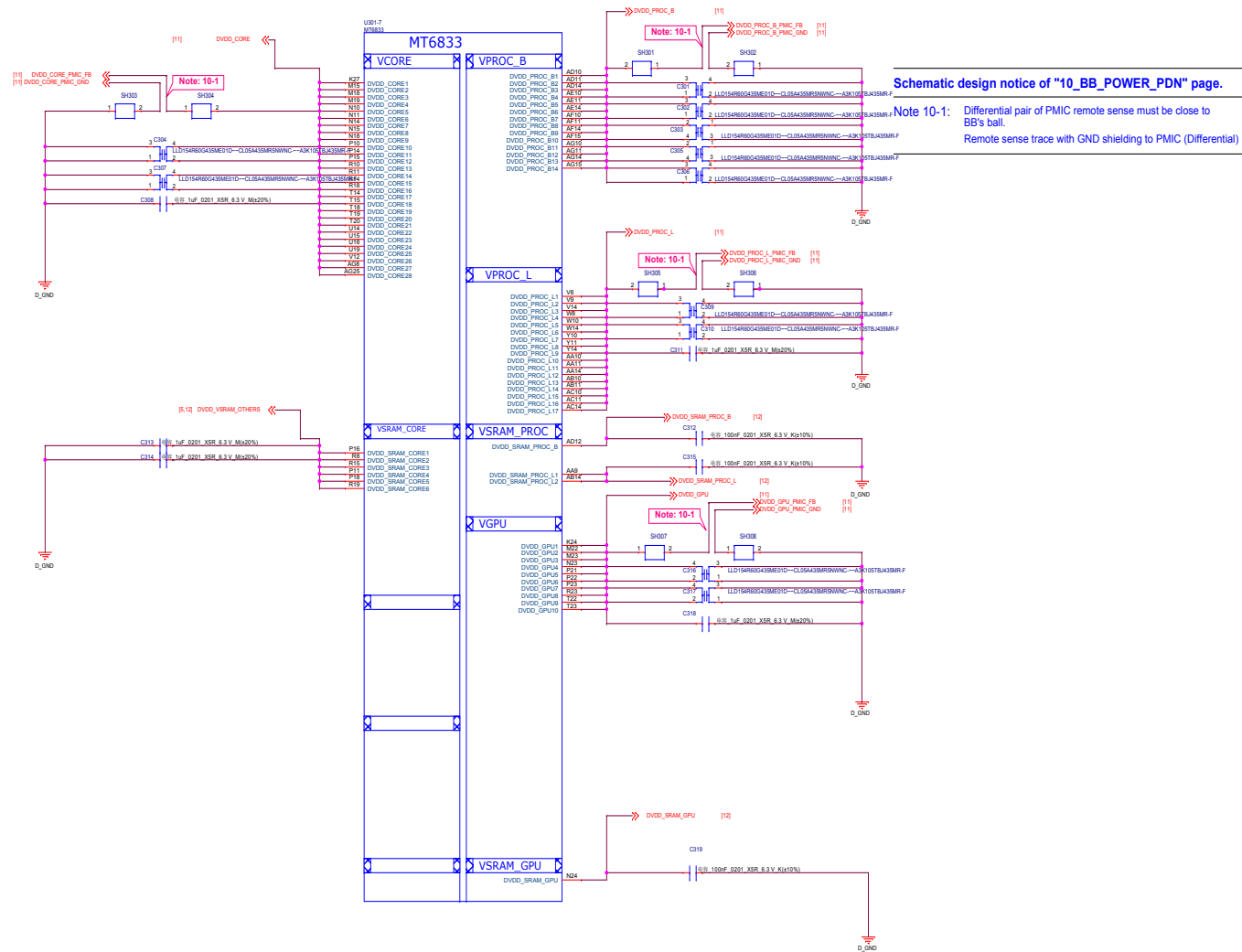
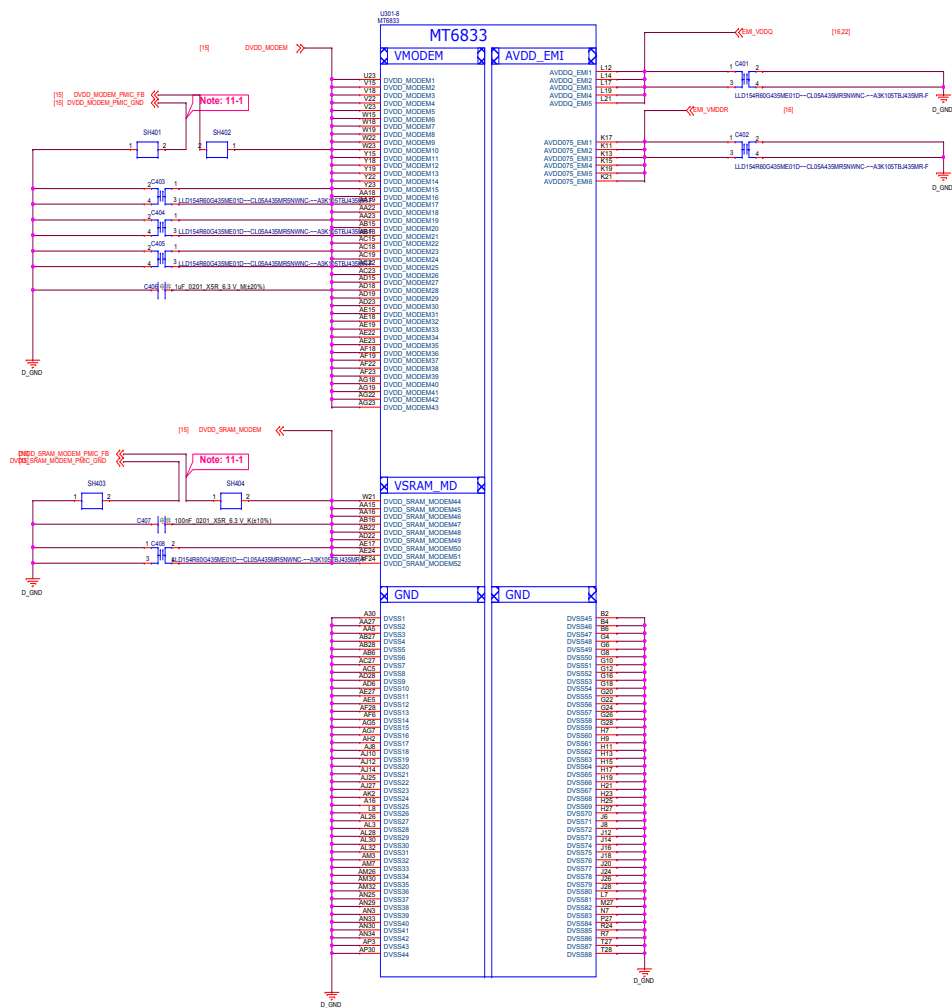


Date	Version	Page	Change notice
2020.06.xx	Pre-Released		
2020.07.xx	V01		
2020.08.14	V02	13_BB_1 29_POWER_EXT 20_POWER_MT6365_Buck 25_POWER_MT6315_VMD	Disconnect Dummy pins A1,A34,AP1,AP34 to GND Change C2923 cap from 6.3V to 10V [PDN] Change Caps of DVDD_GPU - C2039/C2038 to NC [PDN] Change Caps of DVDD_MODEM - C2502/C2534/C2523/C2522 to NC
2020.08.26	V03	File name	Modify the file name

华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title SM6350_Control			
Size D	Project <Doc>		Rev V1
Date: Tuesday, January 12, 2021		Sheet 2 of 48	



华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		03.BB_POWER_PDN1	
Size	Project	<Doc>	Rev
D			V1
Date:	Tuesday, January 12, 2021	Sheet 3 of 48	

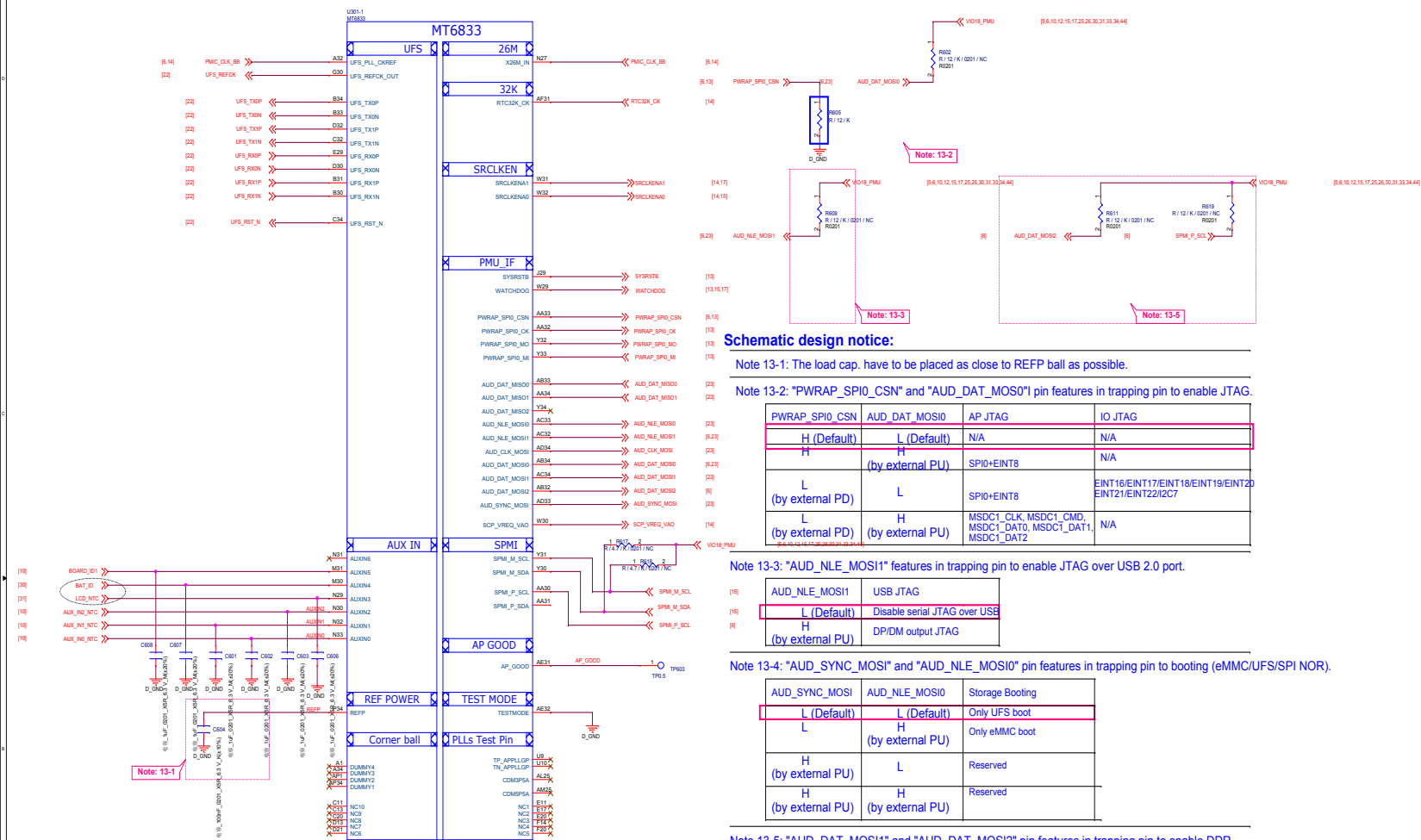


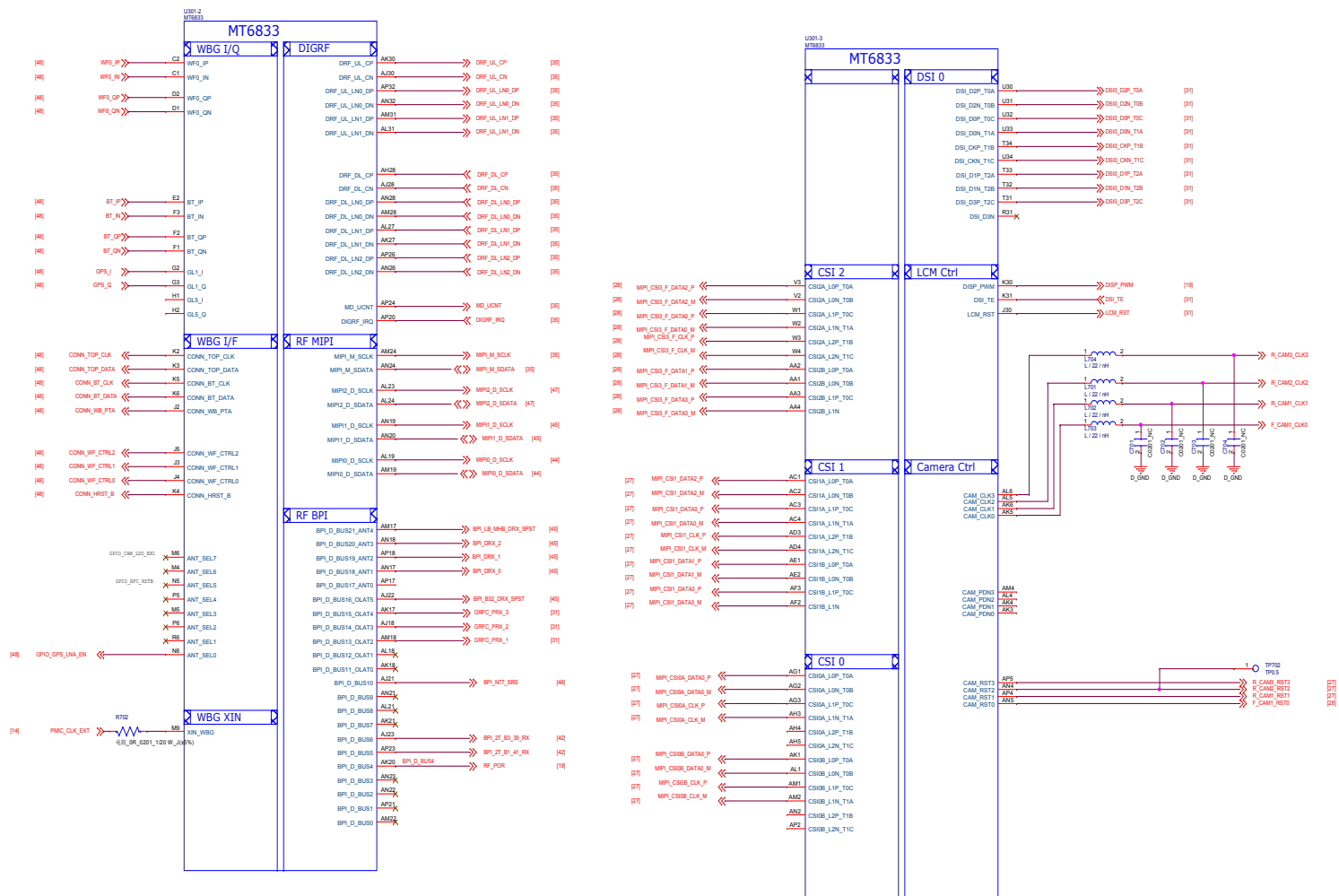
Schematic design notice of "11_BB_POWER_PDN" page.

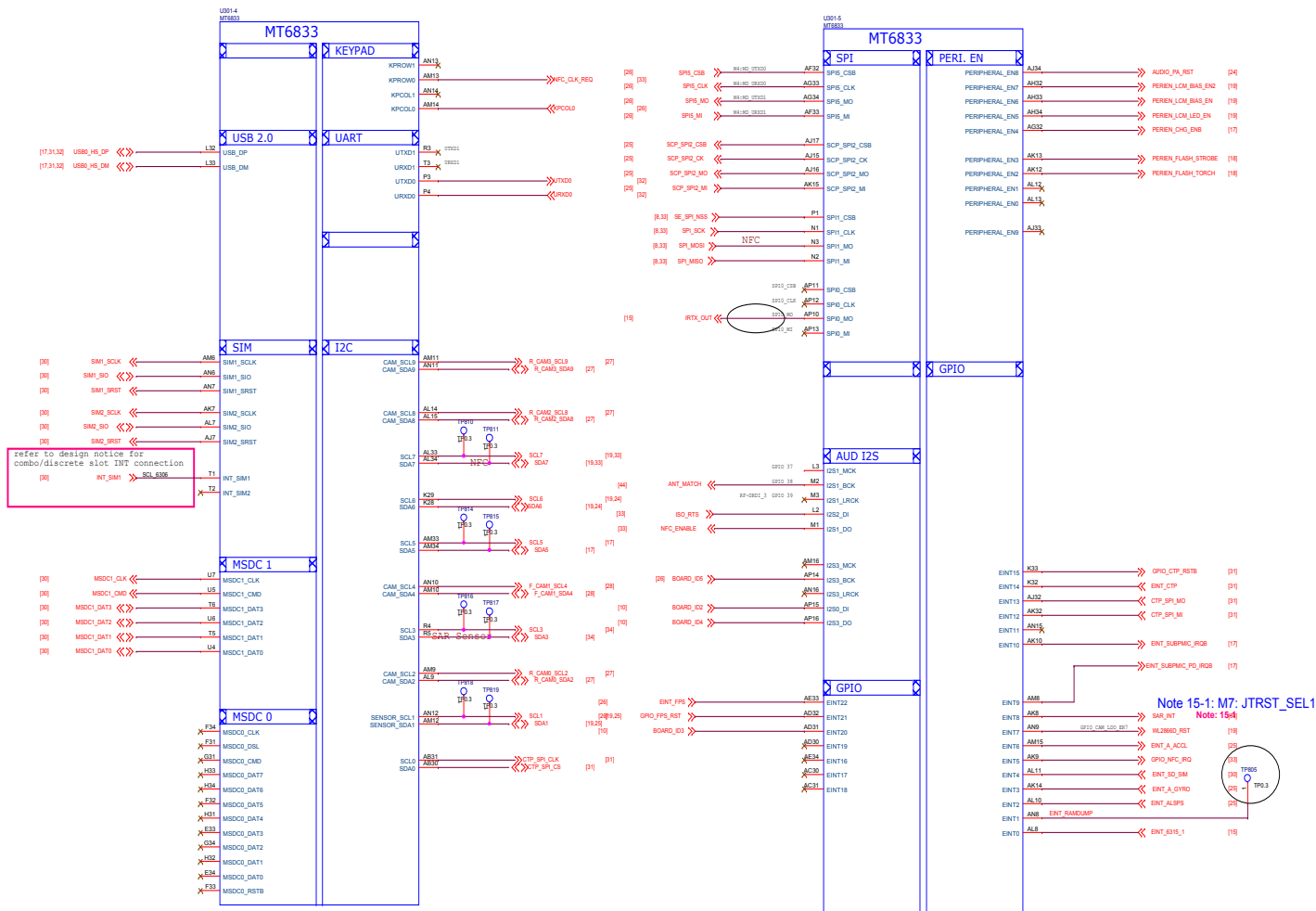
Note 11-1: Differential pair of PMIC remote sense must be close to BB's ball.
Remote sense trace with GND shielding to PMIC (Differential)

华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		04.BB_POWER_PDN2	
Size	Project	<Doc>	Rev
D			V1
Date: Tuesday, January 12, 2021		Sheet 4 of 48	

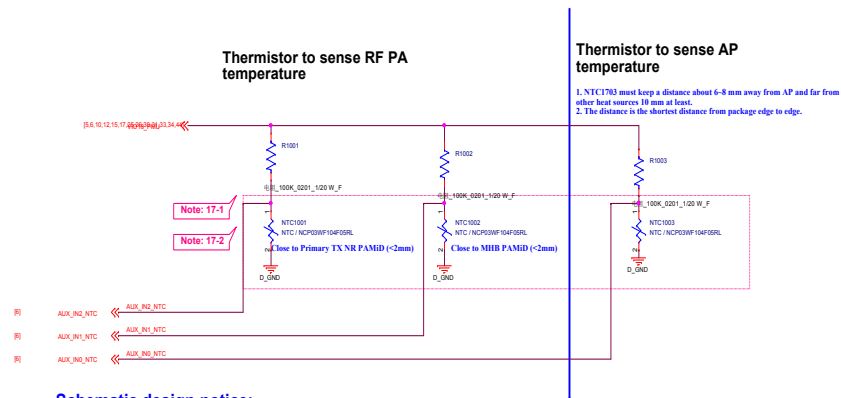








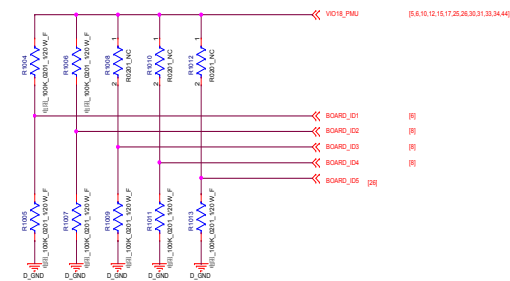
华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		08.BB_3	
Size	Project	<Doc>	Rev
D			v1
Date: Tuesday, January 12, 2021		Sheet 8 of 48	



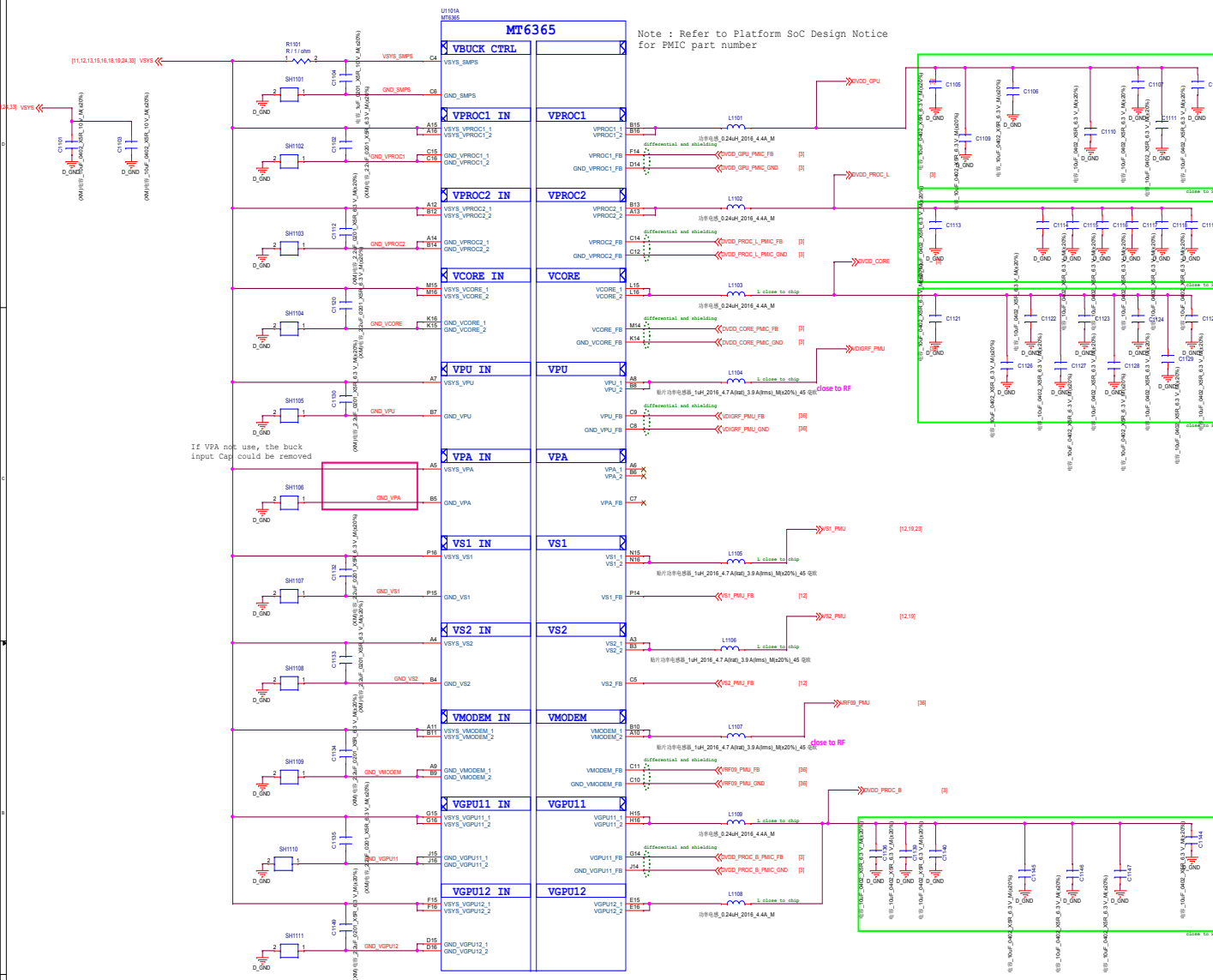
Schematic design notice:

Note 17-1: SW Default Configuration is as followings
AUX_IN0_NTC is for AP Temp.
AUX_IN1_NTC is for MHB PAMID Temp.
AUX_IN2_NTC is for NR PAMID Temp.

Note 17-2: If your design or placement is different with SW Default Config.
Please refer to "MTXXXX Thermal User Manual.docx" and modify SW setting.



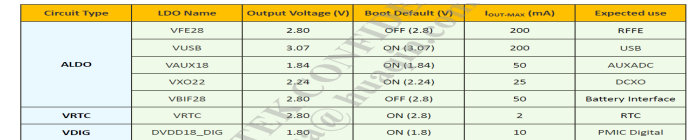
华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		10.BB_AUXADC_Thermal/BOARD_ID	
Size	Project	<Doc>	Rev
D			V1
Date:	Tuesday, January 12, 2021	Sheet	10 of 48



Circuit Type	Buck Name (Application name)	Output Voltage Range (V)	Boot Default (V)	I _{OUT-MAX} (mA)
Buck	VPROC1 (DVDD_GPU)	0.40 ~ 1.19 (6.25mV/step)	ON (0.75)	4800
	VPROC2 (DVDD_PROC_L)	0.40 ~ 1.19 (6.25mV/step)	ON (0.75)	4800
	VGPU11+12 (DVDD_PROC_B)	0.40 ~ 1.19 (6.25mV/step)	ON (0.75)	4800 * 2
	VCORE (DVDD_CORE)	0.40 ~ 1.3 (6.25mV/step)	ON (0.75)	4800
	VMODEM (VRF09_PMU)	0.50 ~ 1.10 (6.25mV/step)	OFF	4800
	VPU (VDIGRF_PMU)	0.40 ~ 1.19 (12.5mV/step)	ON (0.7)	2400
	VS1 (VS1_PMU)	1.86 ~ 2.20 (12.5mV/step)	ON (2.0)	2200
	VS2 (VS2_PMU)	1.20 ~ 1.50 (12.5mV/step)	ON (1.35)	2500
	VPA (VPA_PMU)	0.50 ~ 3.40 (50mV/step)	OFF	1000

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Title		11.POWER_MT6365_Buck	
Size D	Project <Doc>	Rev V1	
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1. "Typical Cap" defined in design notice is the minimum cap. to LDO Cout.
2. NC cap can move to application, if (PCB L<20nH, PCB R<0.2mohm)
=> value and placement of Cap, please refer design notice



Circuit Type	LD0 Name	Output Voltage (V)	Boot Default (V)	I _{out-max} (mA)	Expected use
DLD0	VSIM1	1.71/1.8/1.86/ 2.76/3.0/3.1	OFF (1.86)	200	SIM
	VSIM2	1.71/1.8/1.86/ 2.76/3.0/3.1	OFF (1.86)	200	SIM
	VCM33_1	3.3/3.4/3.5/5.6	OFF (3.3)	800	Connectivity
	VCM33_2	3.3/3.4/3.5/5.6	OFF (3.3)	800	Connectivity
	VEMC	2.55/2.9/3.0/3.3	ON (3.0)	800	eMMC / UFS
	VI02B	2.8/2.9/3.0/ 3.1/3.2/3.3	OFF (2.8)	200	IO & Sensor
	VIBR	2.7/2.8/3.3	OFF (2.8)	200	Vibrator

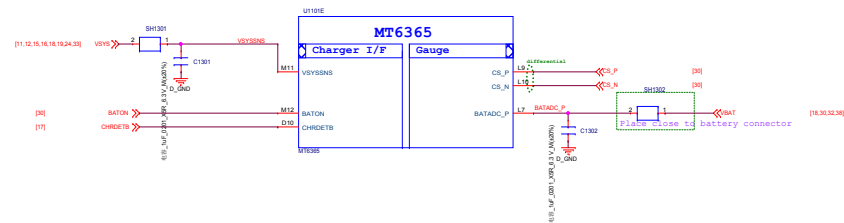
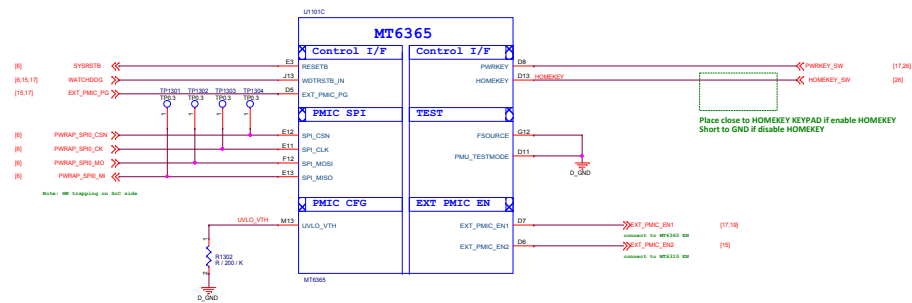
Circuit Type	IDD Feature	Output Voltage (V)	Boost Output (V)	Current (mA)	Expected use
DC/DC	VFUSE	1.80	OFF (1.8)	300	EFUSE
	VAUDIS	1.80	ON (1.8)	300	Audio
	VCAMIO	1.80	OFF (1.8)	300	Camera IO
	VMIB	1.80	ON (1.8)	300	DRAM
	VMFIR	1.80	OFF (1.8)	450	RF
	VIOIR	1.80	ON (1.8)	600	IO & Sensor
	VENIS	1.80	OFF (1.8)	1200	Connectivity
	VIFS	1.85	ON (1.86)	1200	eMMC / UFS
DC/DC LDO	VFCKE	1.80 to 3.00 1.80 to 3.00	ON (1.4)	10	DC/DC
	VBBCK	1.24	ON (1.24)	10	DC/DC

Circuit Type	LDO Name	Output Voltage (V)	Best Default (V)	I _{load_max} (mA)	Expected use
SLDO2	VA09	0.85	ON (0.85)	300	DIGP SRAM (VPFOPBS_MEM_PMU)
	VA12	1.2	ON (1.2)	300	AP Analog Module
	VCN13	1.3	OFF (1.3)	350	Connectivity
	VSRAM_PROC1	0.60 ~ 1.2 (6.25mV/step)	ON (0.85)	600	CPUB SRAM
	VSRAM_PROC2	0.60 ~ 1.2 (6.25mV/step)	ON (0.85)	600	CPUL SRAM
	VSRAM_OTHERS	0.60 ~ 1.2 (6.25mV/step)	ON (0.75)	600	OTHERS SRAM
	VSRAM_MD	0.60 ~ 1.2 (6.25mV/step)	ON (0.85)	600	GPU SRAM
	VRF12	1.2	ON (1.2)	800	AP Analog Module

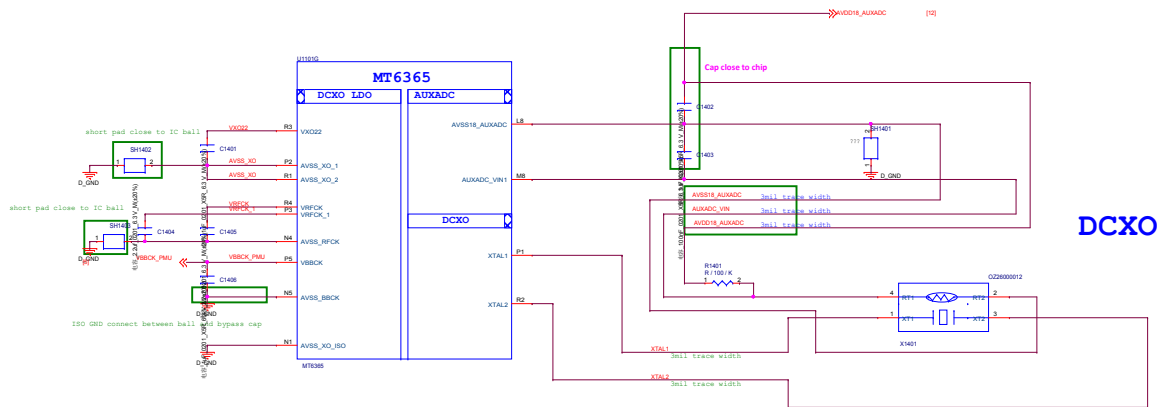
Title	12.POWER_MT6365_LDO
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Rev
V1

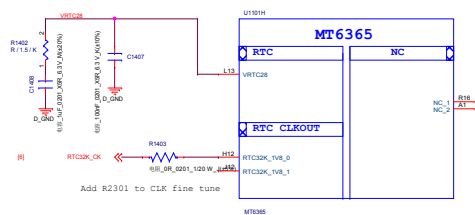
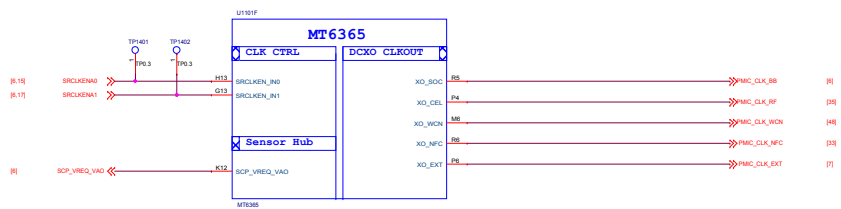
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华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		13.POWER_MT6365_General	
Size	Project	<Doc>	Rev
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Route XTAL1/XTAL2/ AVDD18_AUXADC/ AUXADC_VIN/ AVSS18_AUXADC with 3mils width traces and well GND shielding.



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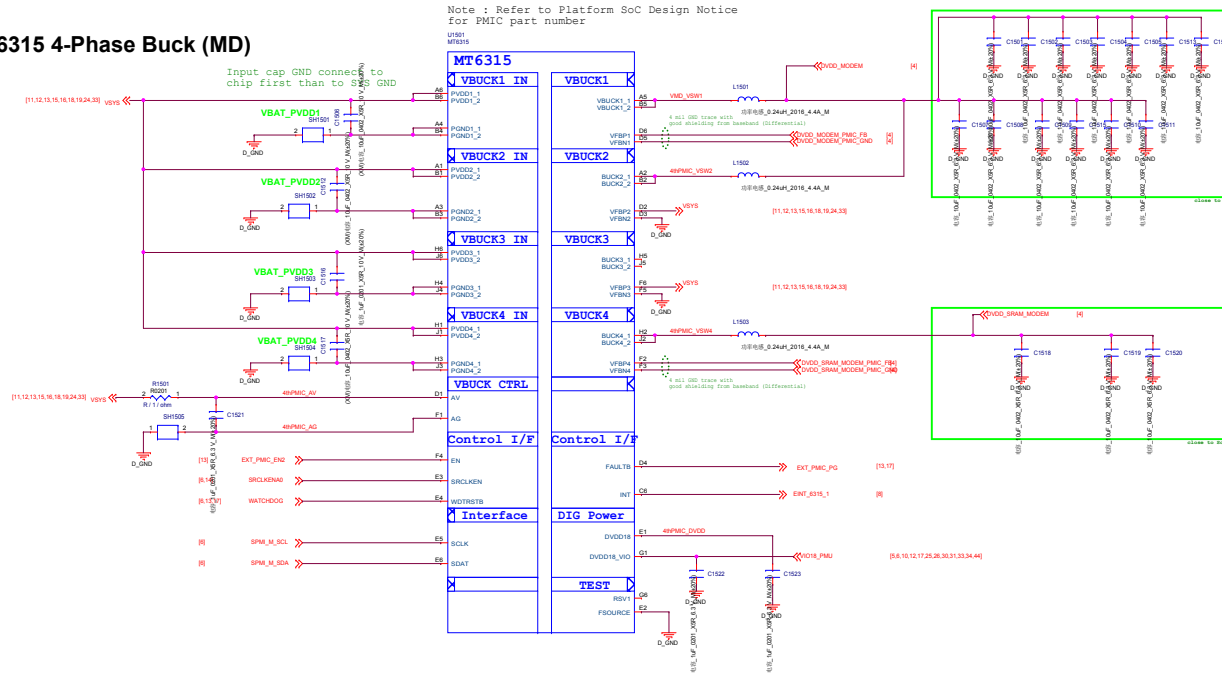
Title 14.POWER_MT6365_Clock

Size D Project <Doc> Rev V1

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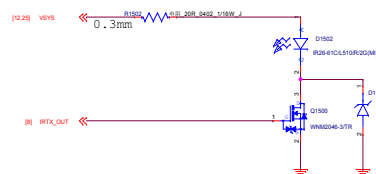
MT6315 4-Phase Buck (MD)

Note : Refer to Platform SoC Design Notice
for PMIC part number

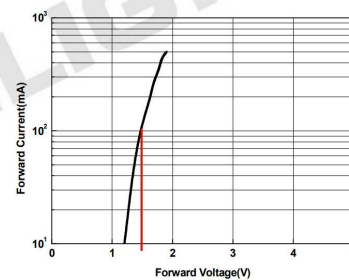


MT6315NP/B	Output Voltage Range (V)	Boot Default (V)	I _{OUT-MAX} (mA)	Expected use
BUCK1+2	0.40 ~ 1.19375 (6.25mV/step)	ON (0.825)	5000*2	DVDD_MODEM
BUCK3	0.40 ~ 1.19375 (6.25mV/step)	ON (0.75)	5000	--
BUCK4	0.40 ~ 1.19375 (6.25mV/step)	ON (0.825)	5000	DVDD_SRAM_MODEM

IR LED



Forward Current vs. Forward Voltage



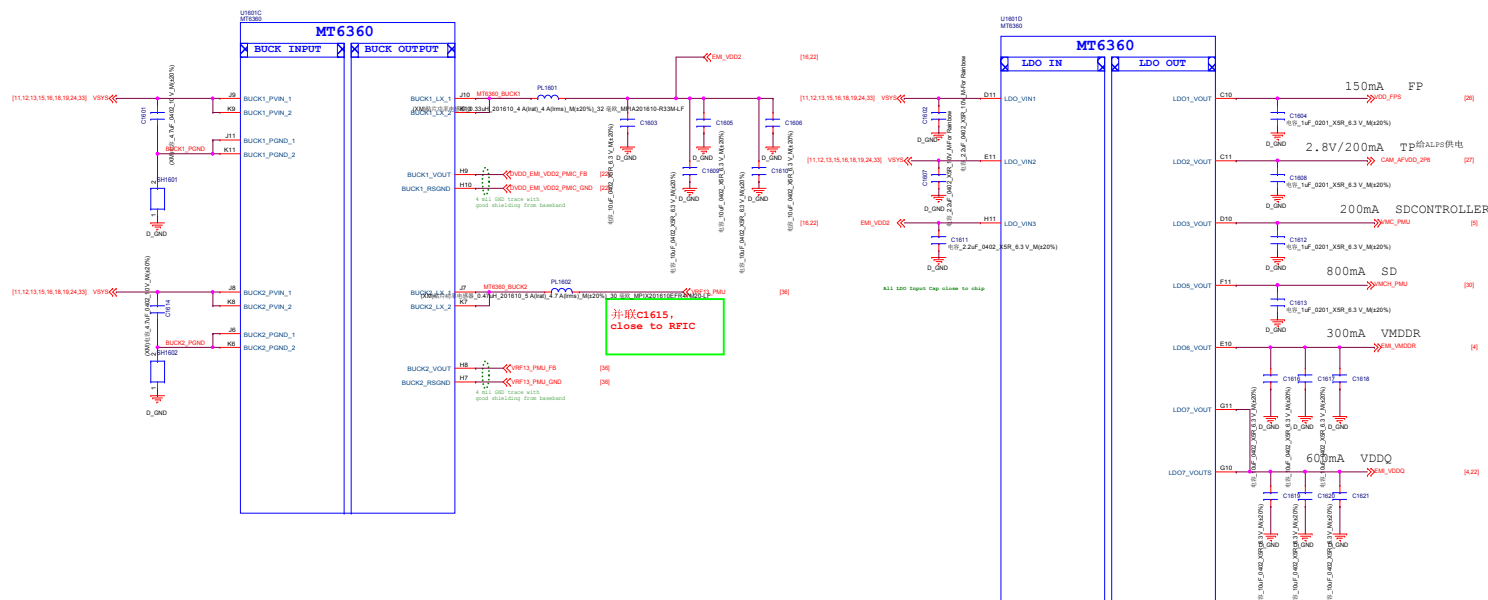
华勤通讯 Huaqin Telecom Technology Com.,Ltd

Title	15.POWER_MT6315_VMD/IR_LED
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Size	Project	Rev
D	<Doc>	v1

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Note : Refer to Platform SoC Design Notice
for PMIC part number

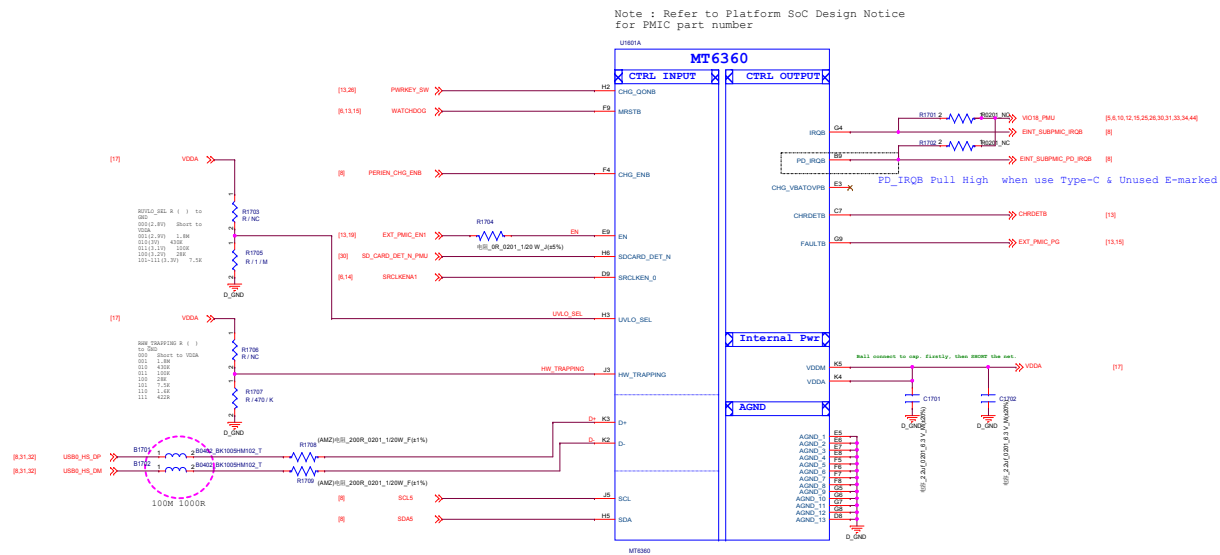


LDO1 VFP
LDO2 VTP
LDO3 VMC
LDO5 VMCH
LDO6 EMI_VMDDR
LDO7 EMI_VDDQ

Circuit Type	Name	Output Voltage Range (V)	Boot Default (V)	I _{OUT-MAX} (mA)	Expected use
Buck	BUCK1	0.3 ~ 1.3 (5mV/step)	ON (1.125)	3000	EMI_VDD2
	BUCK2	0.3 ~ 1.3 (5mV/step)	OFF (1.3)	3000	VRF13_PMU

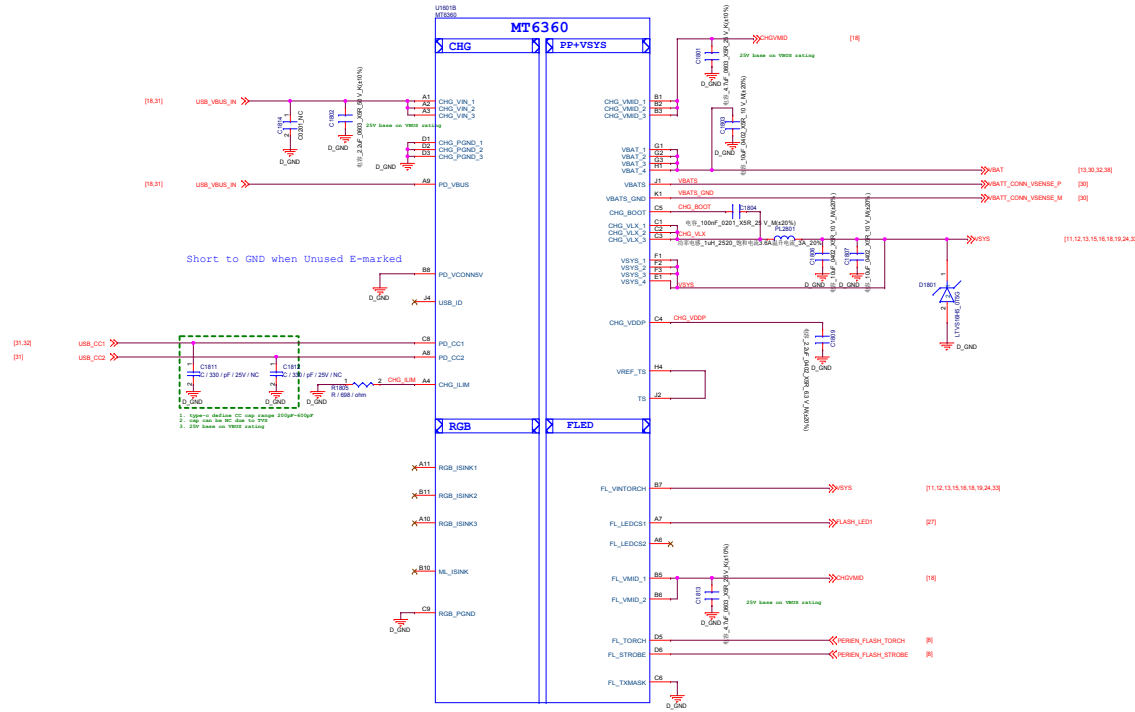
Circuit Type	Name	Output Voltage (V)	Boot Default (V)	I _{OUT-MAX} (mA)	Expected use
LDO	LDO1	1.8/2.0/2.1/2.5/2.7/ 2.8/2.9/3.0/3.1/3.3	OFF (1.8)	150	Fingerprint
	LDO2	1.8/2.0/2.1/2.5/2.7/ 2.8/2.9/3.0/3.1/3.3	OFF (1.8)	200	Touch Panel
	LDO3	1.8/2.9/2.8/3.0/3.3	OFF (3.0)	200	SD Card
	LDO5	2.9/3.0/3.3	OFF (3.0)	800	SD Card
	LDO6	0.75	ON (0.75)	300	EMI_VMDDR
	LDO7	0.6	ON (0.6)	600	EMI_VDDQ

华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 16.POWER_MT6360_BUCK_LDO*		
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Title 17.POWER_MT6360_General			
Size D	Project	<Doc>	
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Note : Refer to Platform SoC Design Notice
for PMIC part number



Schematic design notice of "28_POWER_MT6360_Charger" page.

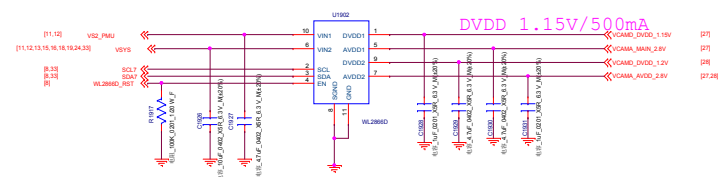
Note 28-1: For better ESD & surge performance we need choose suitable device for system protection.
Please refer to the latest version of [Surge device selection guide] provided by MTK.

华勤通讯 Huaqin Telecom Technology Com.,Ltd

Title 18.POWER_MT6360_Charger

Size D	Project <Doc>	Rev v1
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[illegible][illegible]

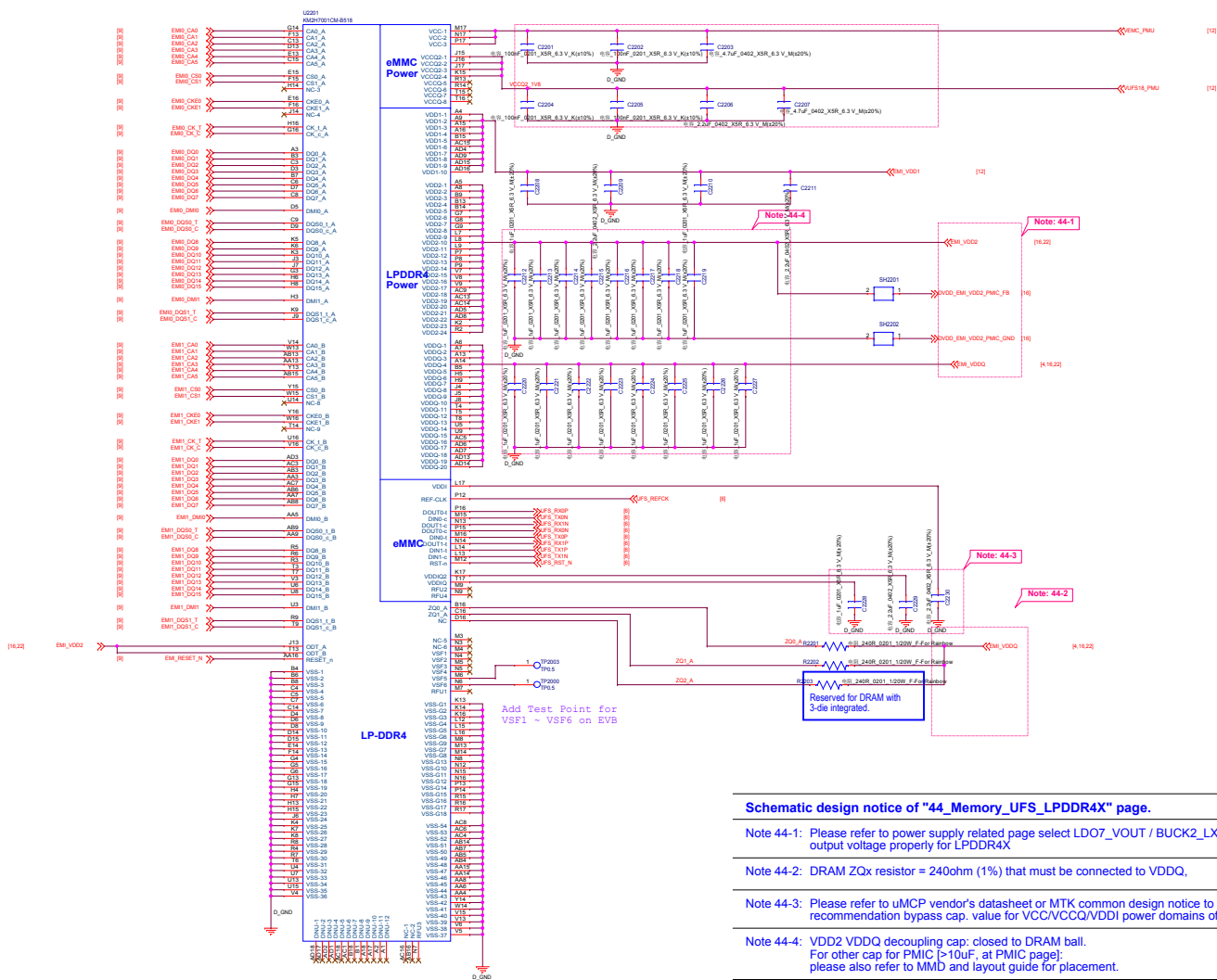
厂家	I2C ADDRESS
WL2866D	0x50A
ST5904	0x50A

厂家	INC ADDRESS
OC92131	0111110 binary (or 3Eh)
9M5109	0111110 binary (or 3Eh)

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Title 19.POWER_EXT			
Size D	Project <Doc>		Rev V1
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Title PM7250B_CHARGE/FG		
Size E	Project <Doc>	Rev v1
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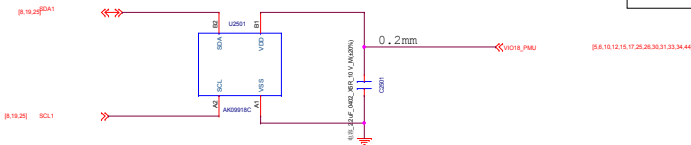
华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title PM7250B_MISC/GPIO/GND		
Size D	Project <Doc>	Rev v1
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华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title			
22.Memory_UFS_LPDDR4X			
Size	Project		Rev
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M-Sensor

厂家	I2C ADDRESS
AK8963C	0ch
AK8963C	2ch

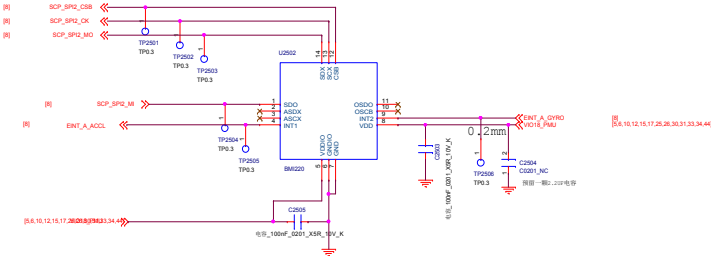


A+Gsensor

default SPI (if sensor hub support)

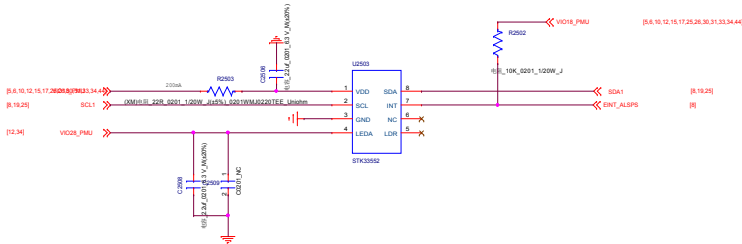
厂家	I2C ADDRESS
BM220	
BM220	

Gyro+A sensor



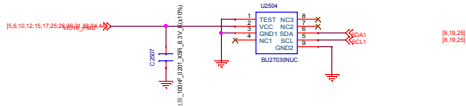
ALS+PS

厂家	I2C ADDRESS
STK3352	0ch



后光感 BU27030NUC

厂家	I2C ADDRESS
BU27030NUC	0ch



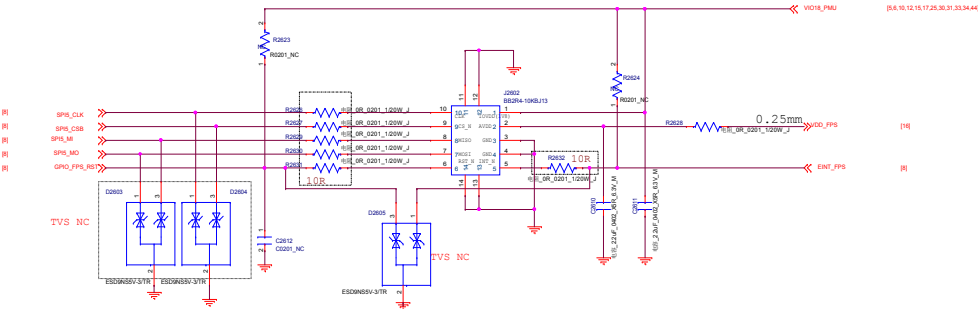
华勤通讯 Huaqin Telecom Technology Com.,Ltd

Title 25.SENSOR

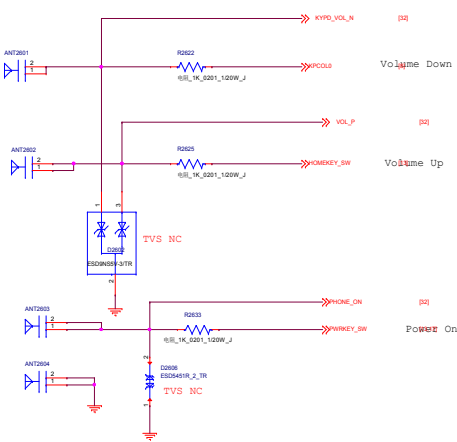
Size D Project <Doc> Rev v1

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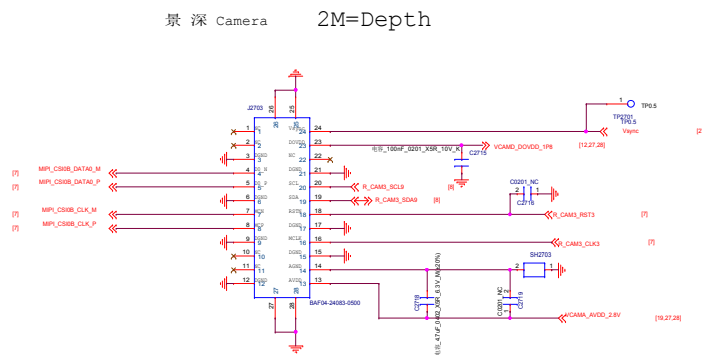
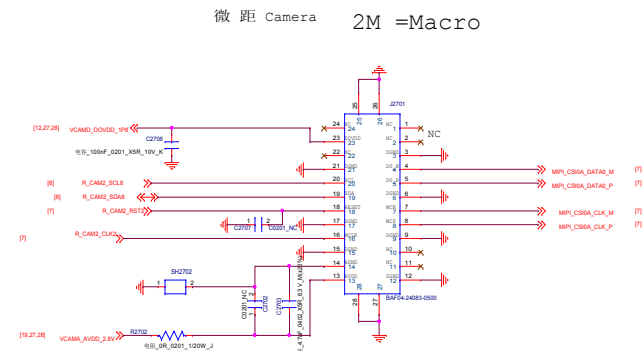
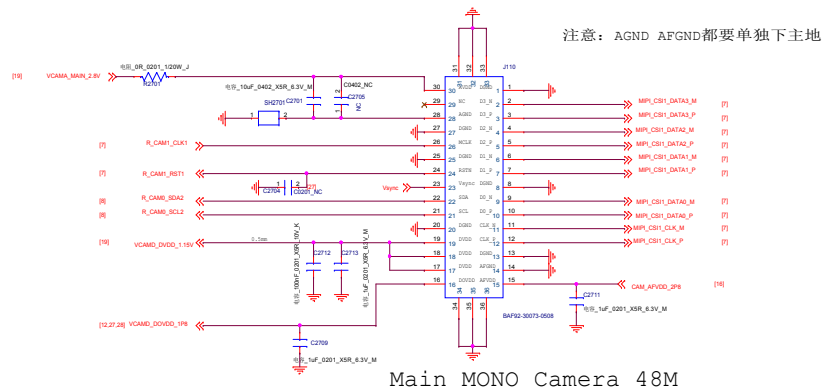
Finger Print



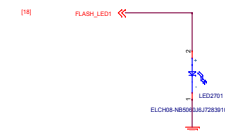
KEY BTB



华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		26.FP/KEY	
Size	Project	<Doc>	Rev
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Flash LED 5V Boost



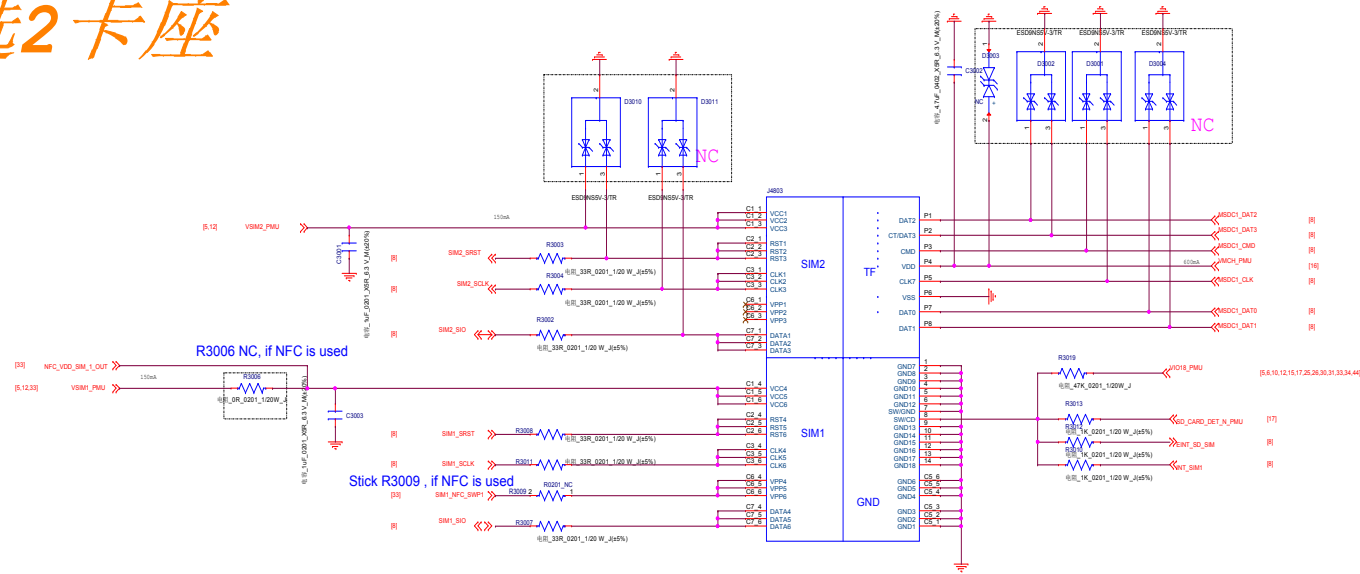
Flash 下方地要完整且足够大，散热一定要好

华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title 27.REAR_CAM			
Size D	Project <Doc>		Rev v1
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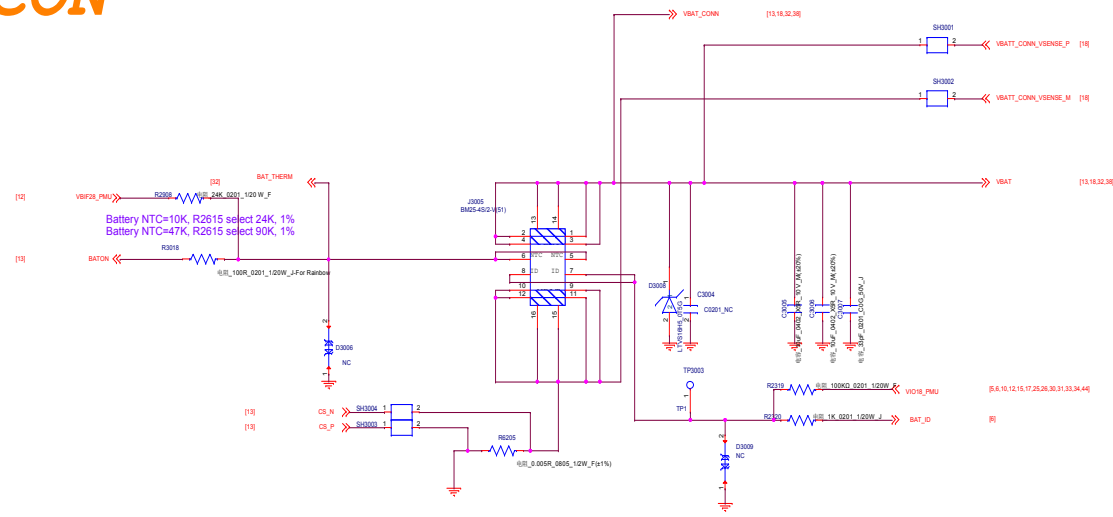
华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title 28.FRONT_CAM			
Size D	Project <Doc>		Rev V1
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Title Sub_Charger_SMB1396		
Size D	Project <Doc>	Rev V1
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3选2卡座



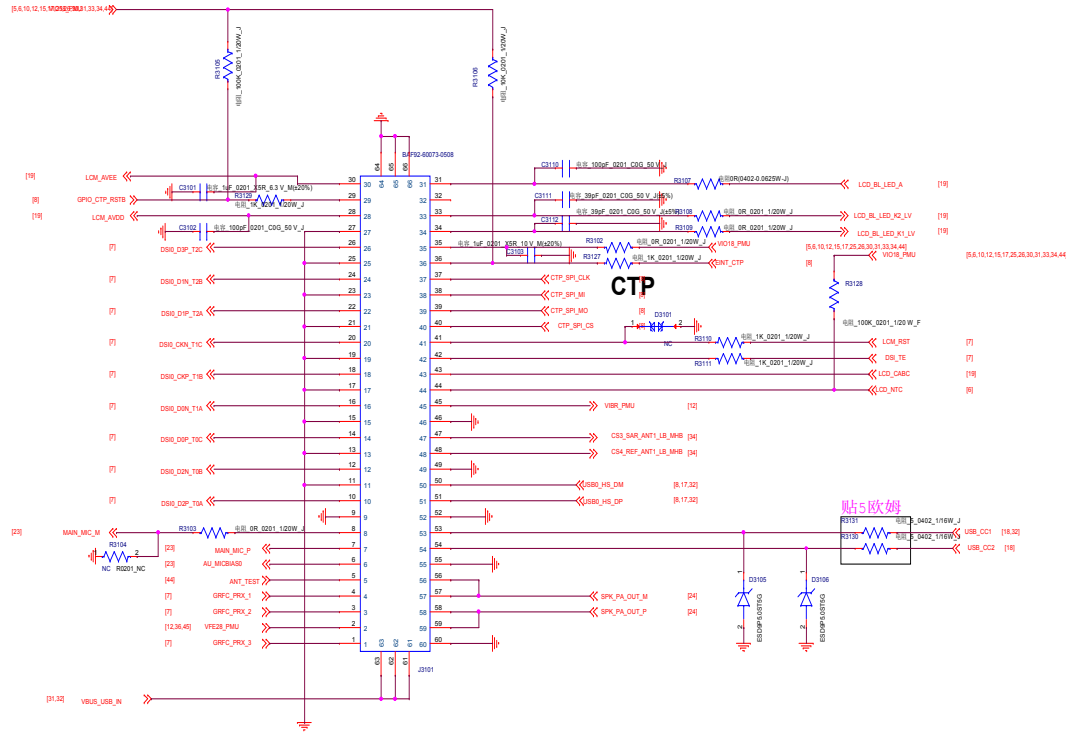
BATT CON



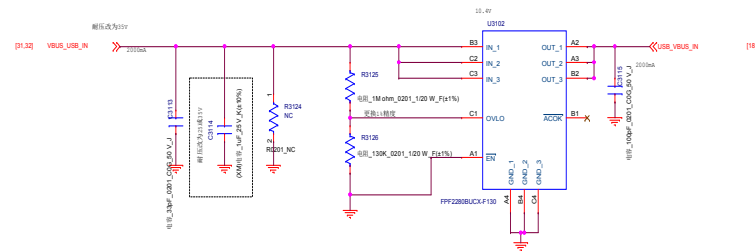
华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		30.SIM/BATT	
Size	Project	<Doc>	Rev
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LINK

0.25mm

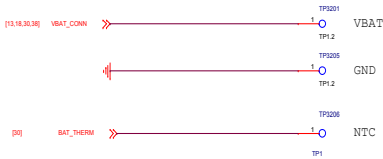


OVP



华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		31.60PIN_LCD_LINK/OVP	
Size D	Project <Doc>		Rev v1
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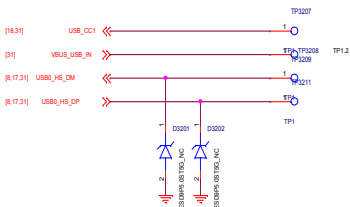
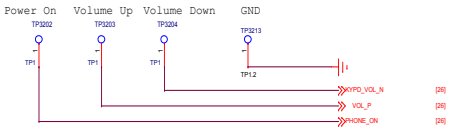
VBAT TEST POINT



VBAT

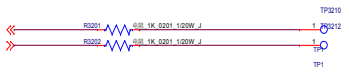
KEYPAD

KEY BTB

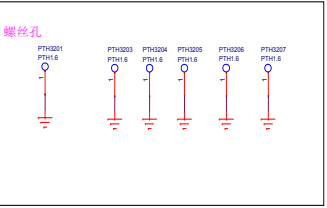
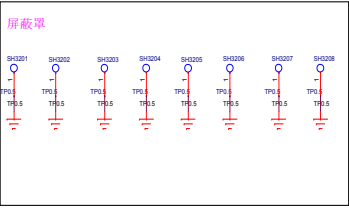


VCHG

UART TP

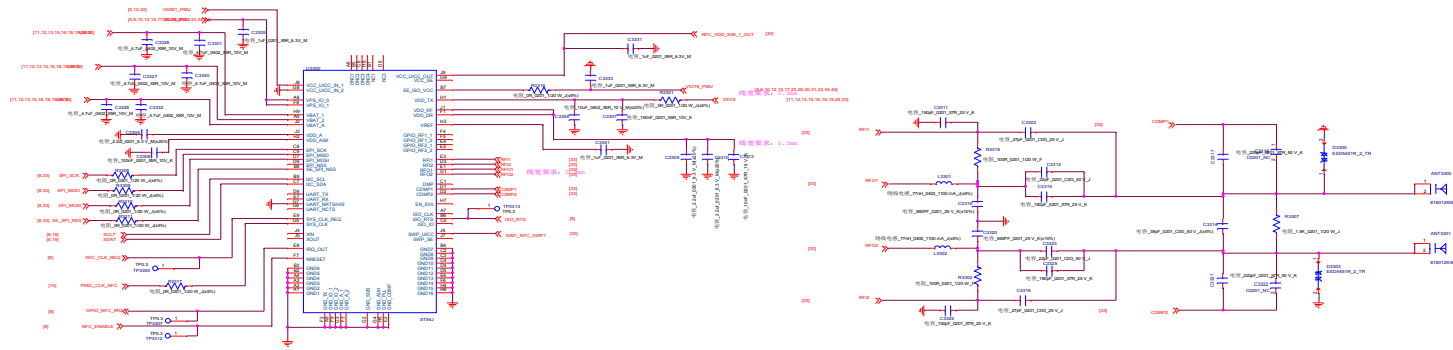


JTAG

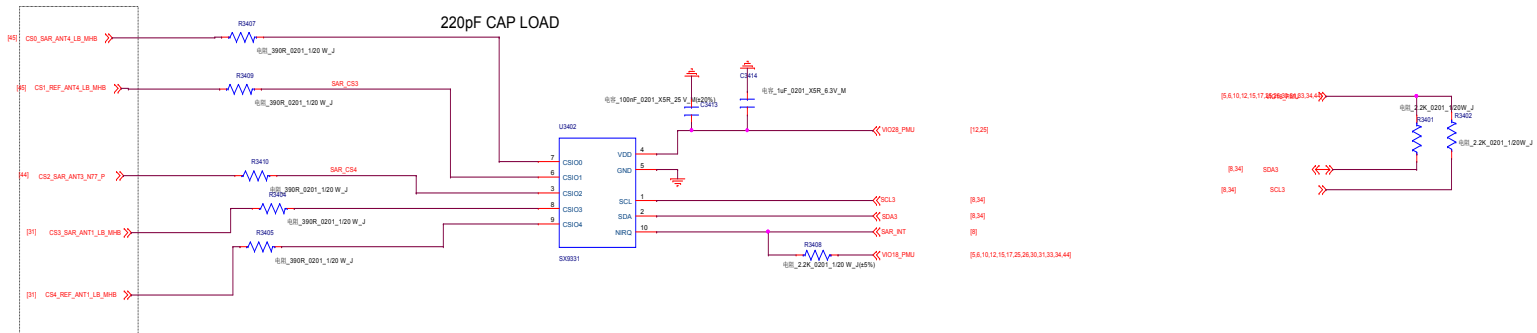


华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		32.TEST_POINT/SHIELD	
Size	Project	<Doc>	Rev
D			V1
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RF_NFC_not support



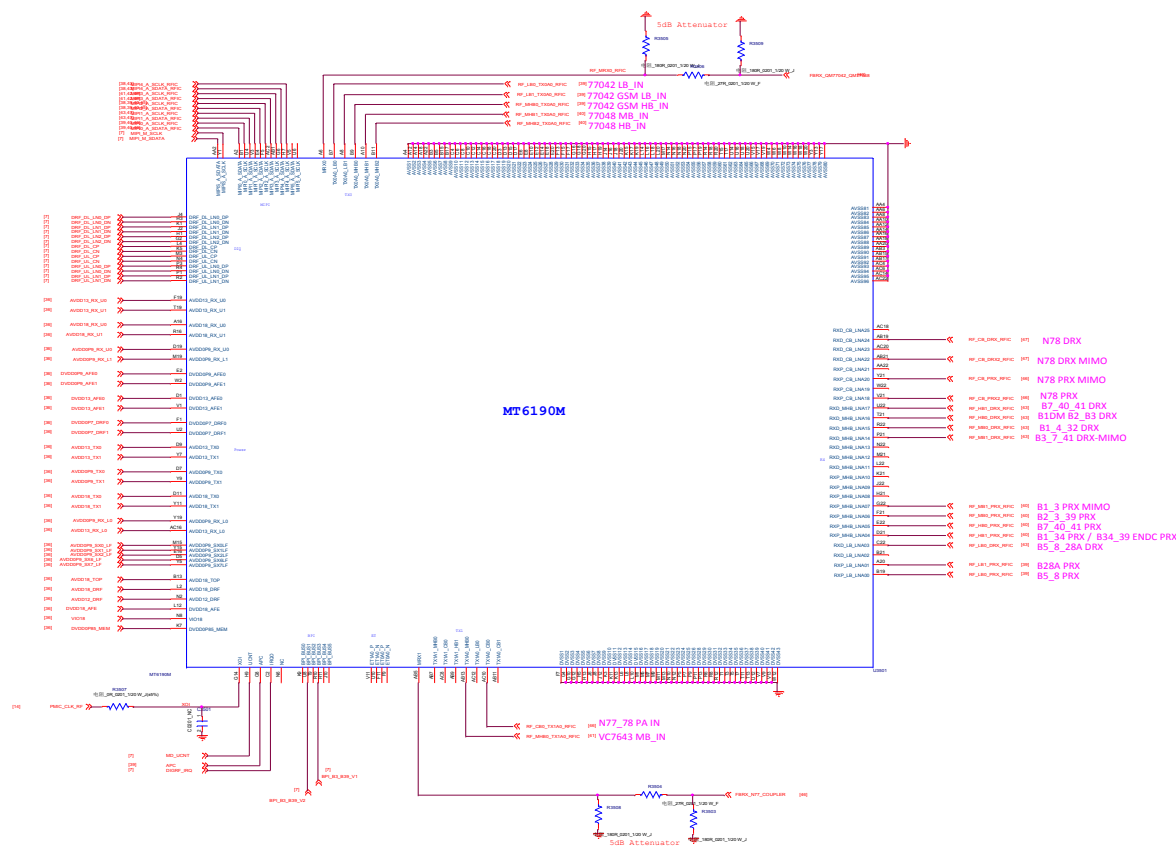
SAR_sensor Not support



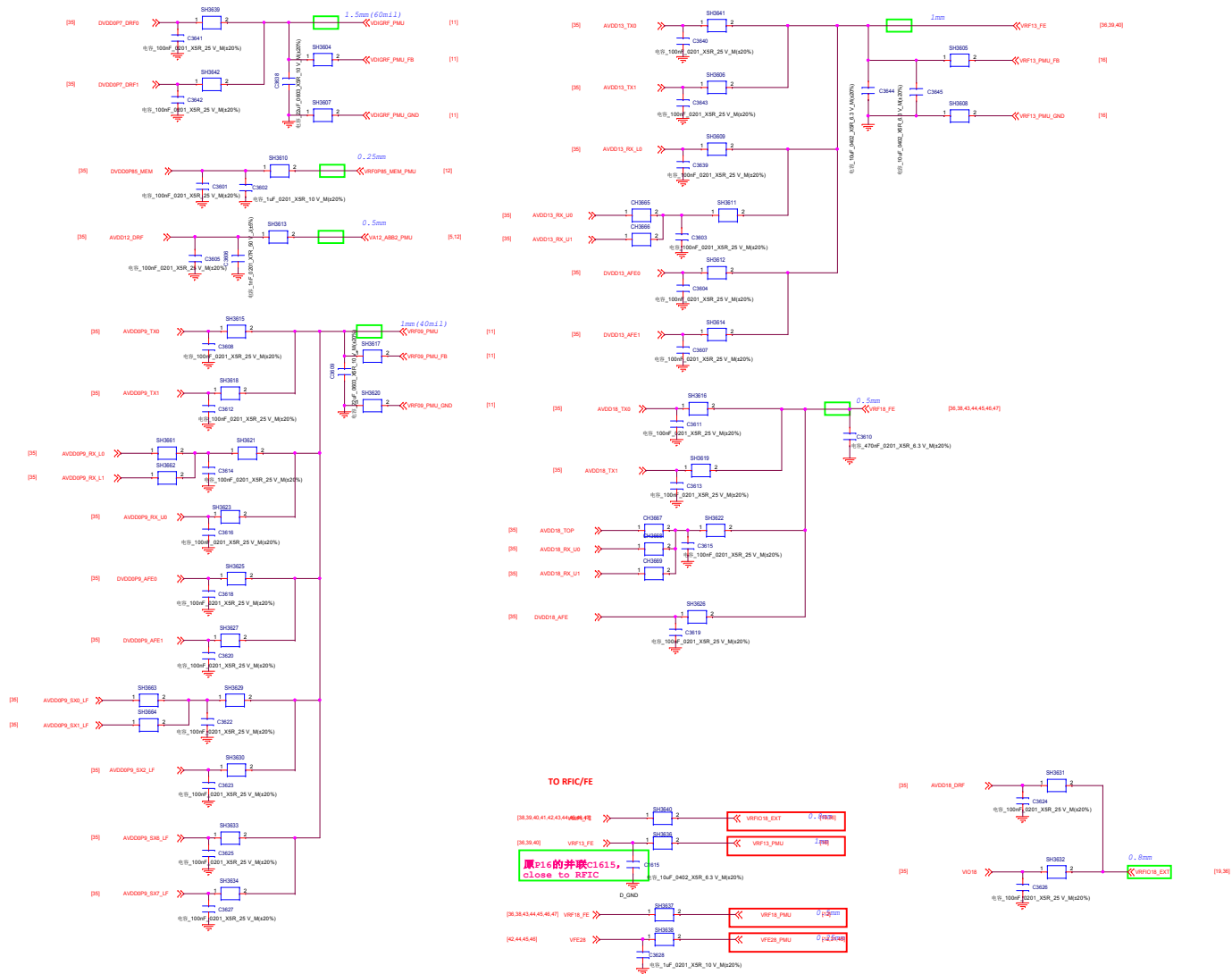
Design notice:

- 1) CS0_1 nets are differential pairs;
- 2) The width for each line, should be less than 0.1mm;
- 3) 0.2mm should be guaranteed between the ground and differential pairs;
- 4) And also, the distance between the differential line should be more than 0.2mm;
- 5) All the CSx nets should be protected by surrounding GND;

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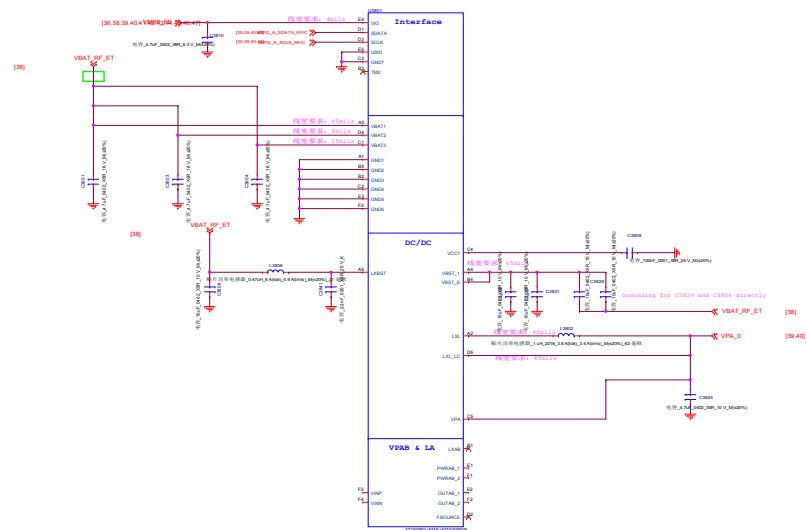
MT6190M_PWR



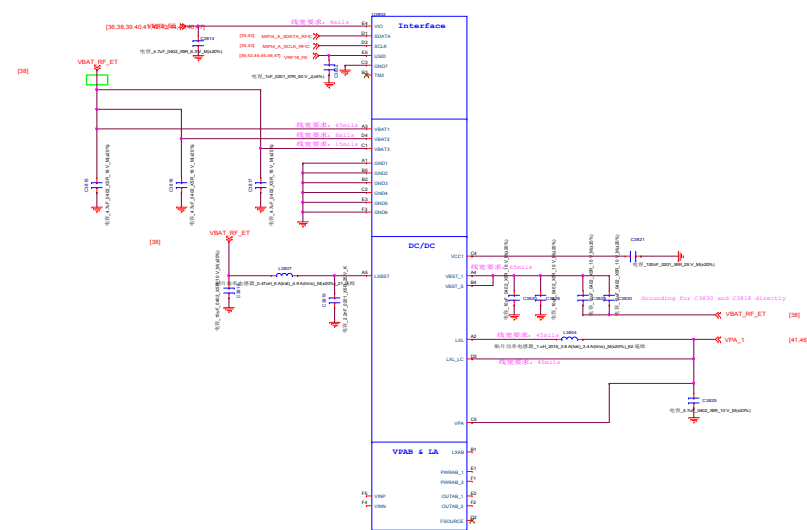
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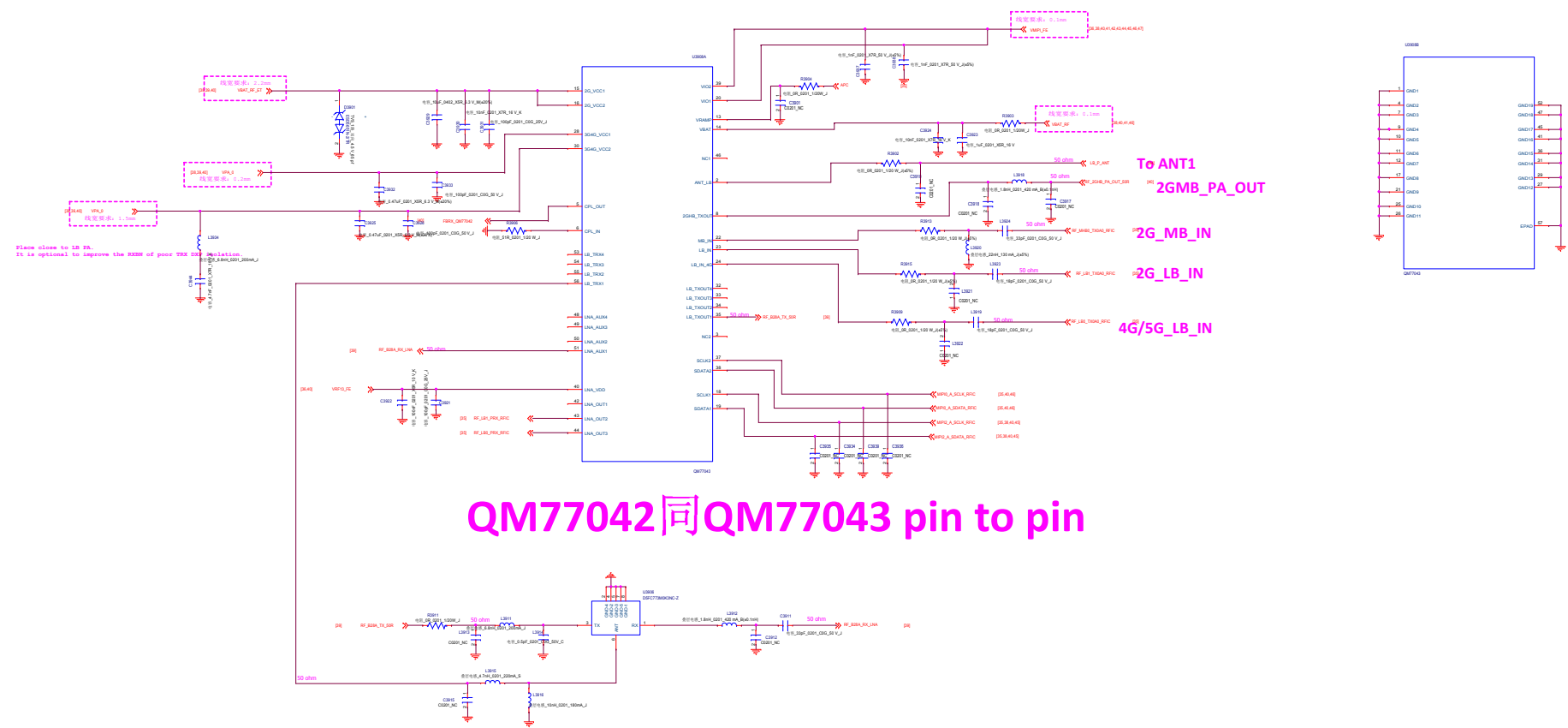
QM77042 and QM77048 for one APT



VC7643 and QM78200 for the other APT



2G_3G/4G/5G LB PA_QM77042

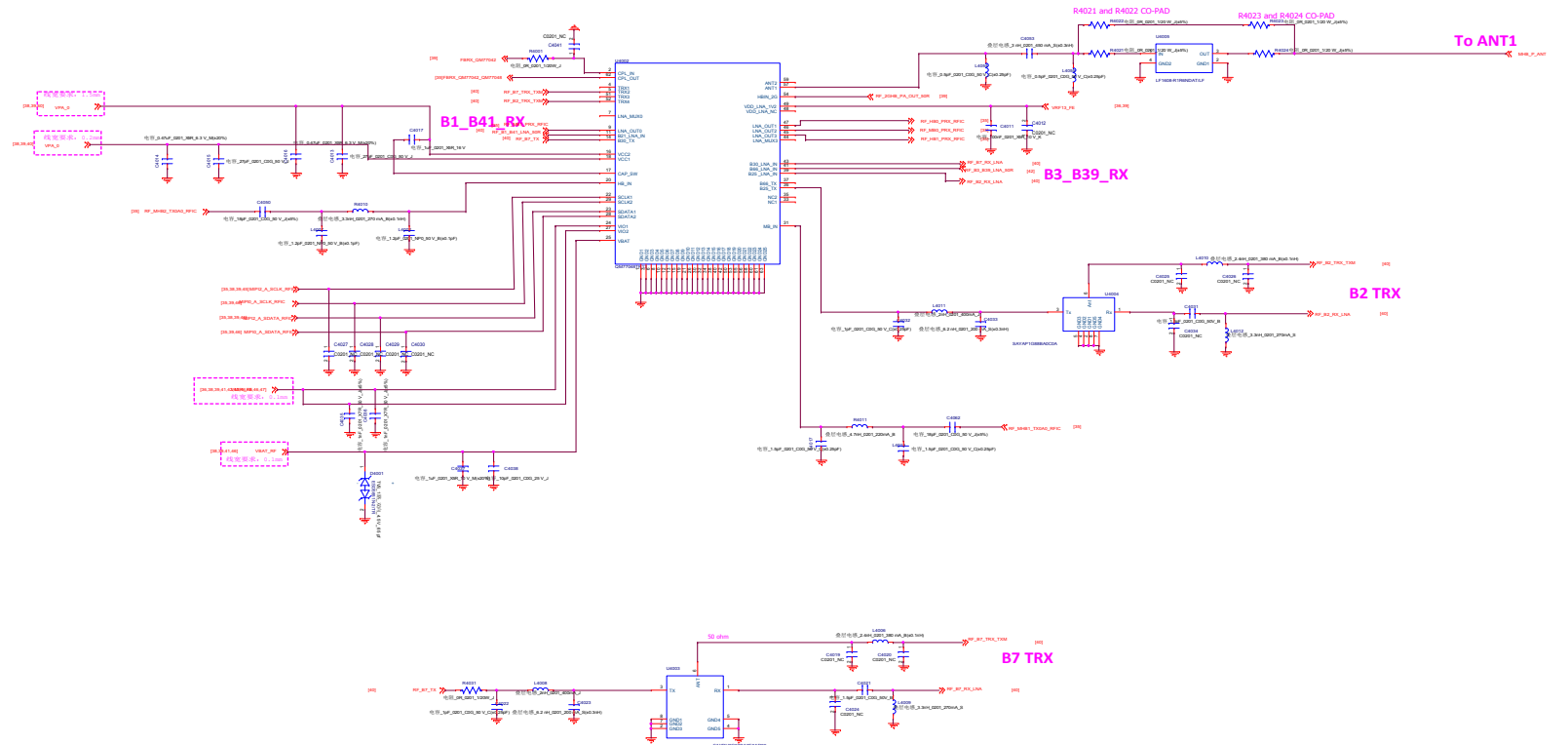


QM77042同QM77043 pin to pin

NTC温度检测
见P10的R1001 and NTC1001

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QM77042			
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4G/5G MHB PA_QM77048E

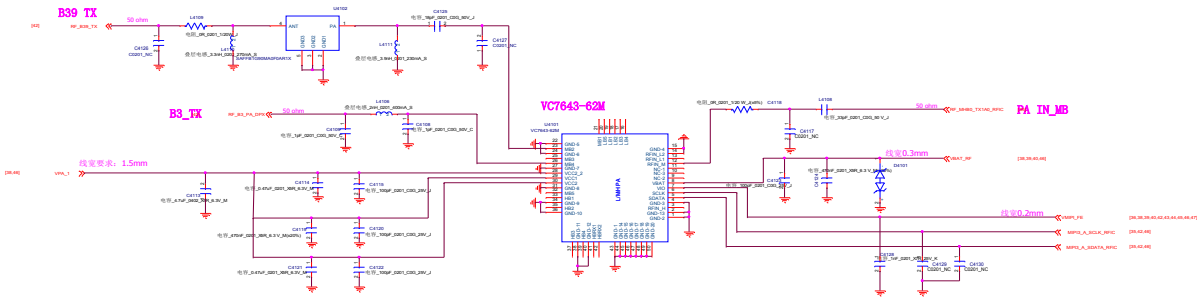


NTC温度检测

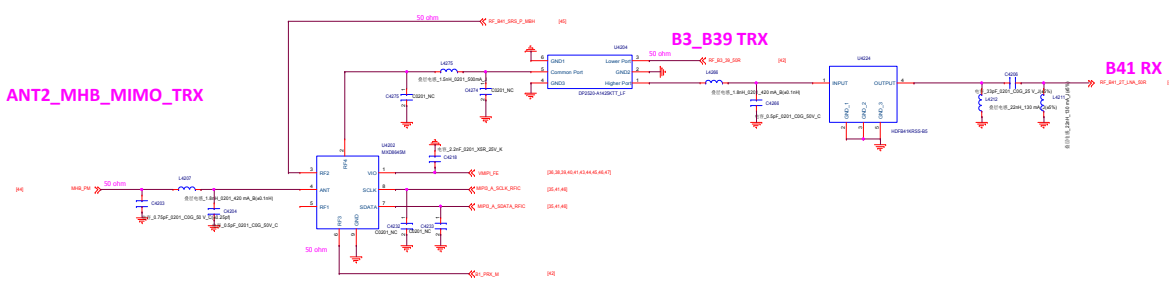
见P10的R1002 and NTC1002

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MMMB PA_MB ENDC



MMMB PA_VC7643_ENDC_MHB_TRX



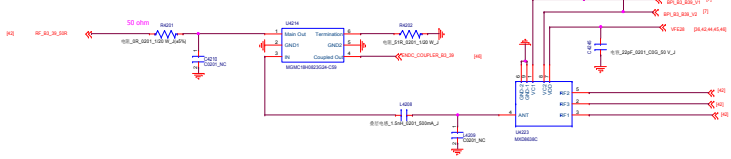
B1_B41_RX



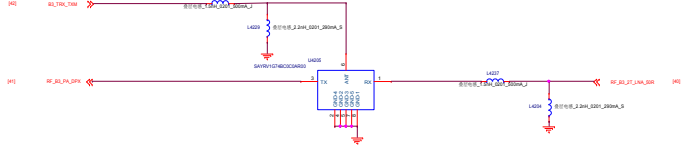
B3_B39_RX



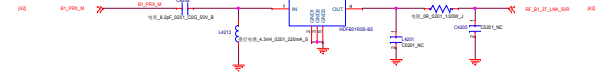
B3_B39(ENDC)



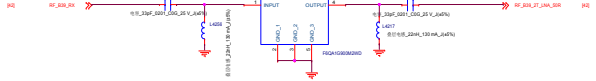
B3 TRX



B1 PM RX

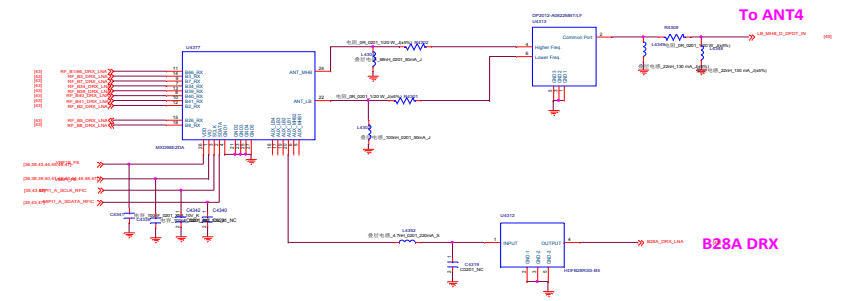
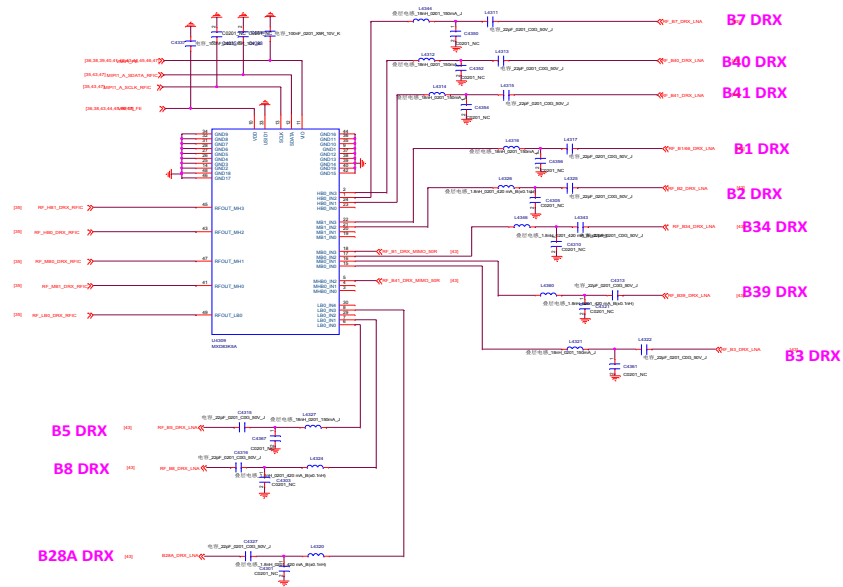


B39 RX

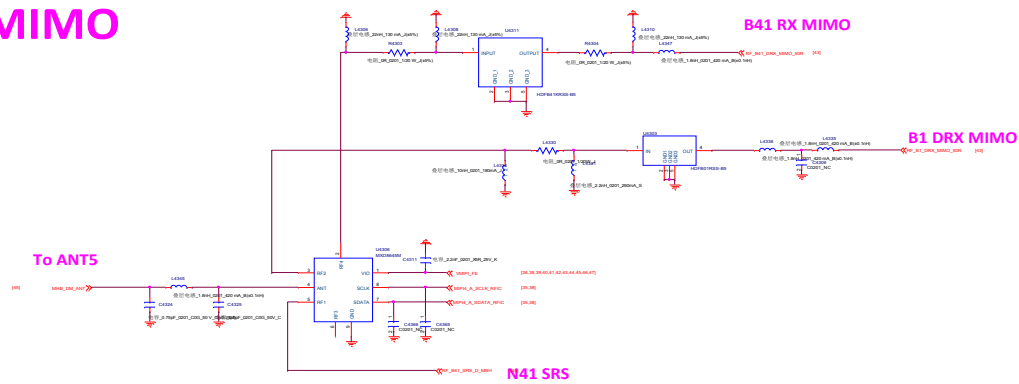


华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title MMMB PA_VC7643_ENDC_MHB_TRX			
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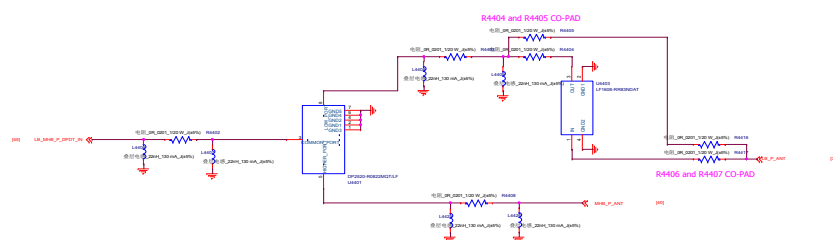
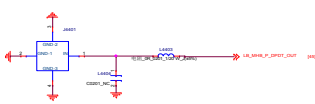
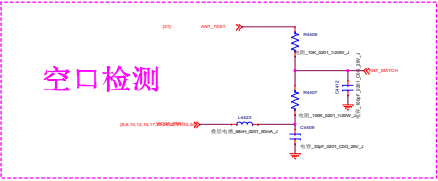
DRX_MXD98E2DA+MXD83K5A



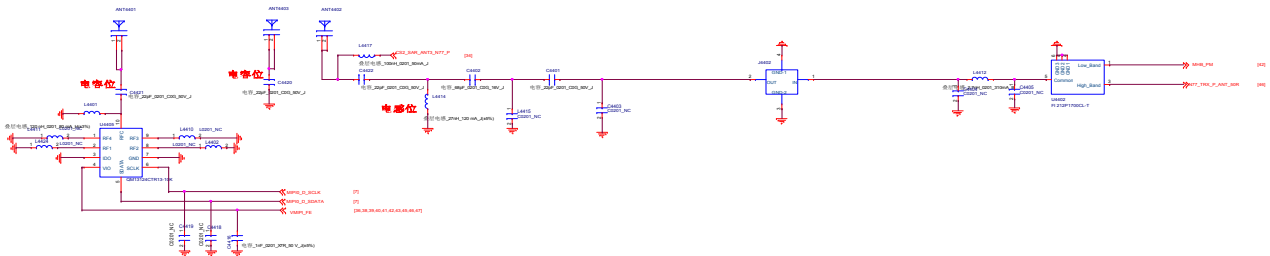
DRX_MIMO



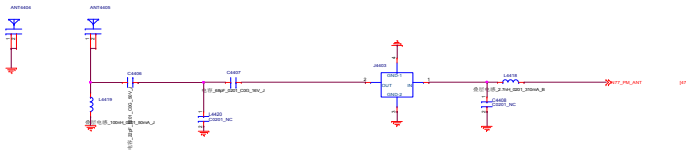
ANT1 (LB_MHB_P) to 小板



ANT2 (MHB_PM/N77_TRX)

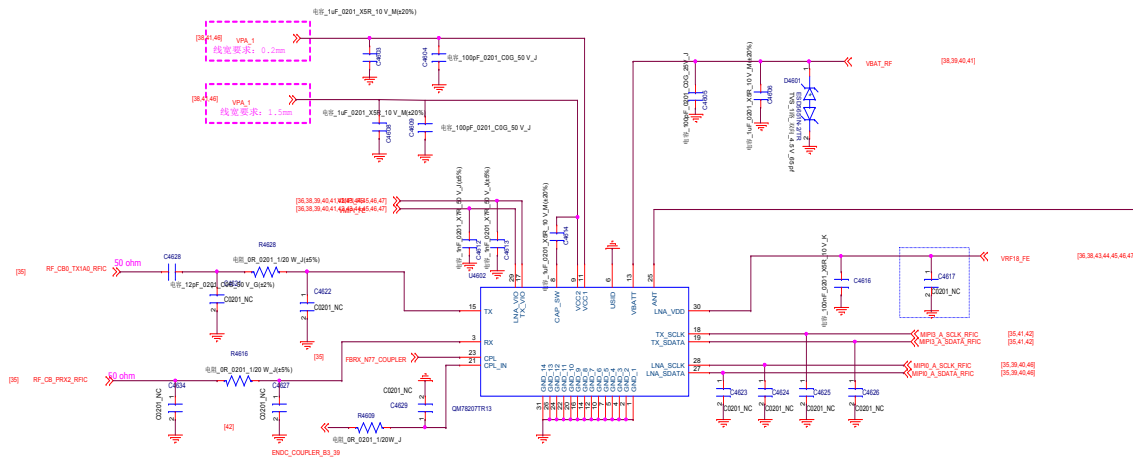


ANT3 (N77_PM)



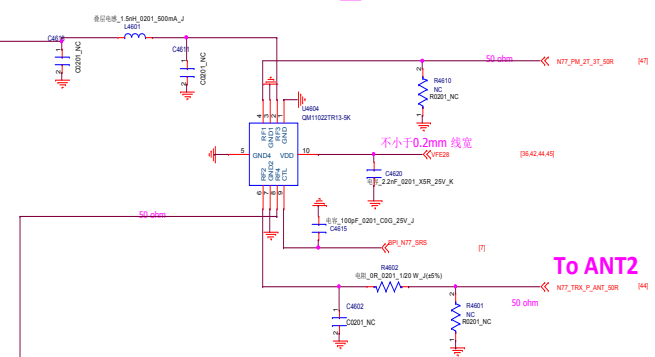
N77/78_TRX0_QM78207

NTC温度检测
见P10的R1002 and NTC1002

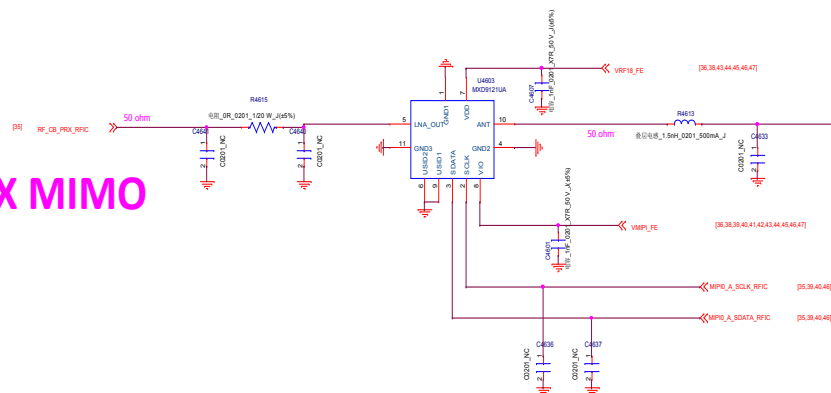


N77_78 PRX

N77/78_SRS



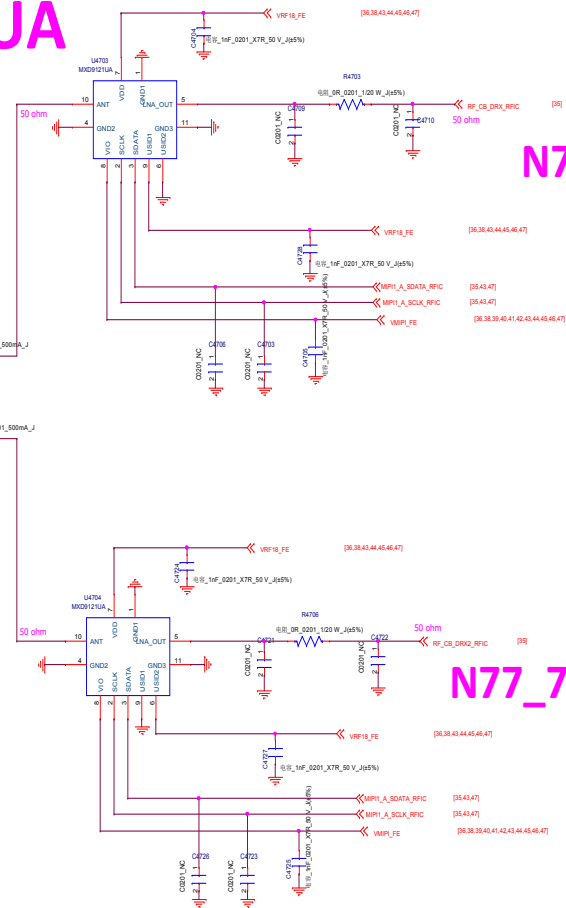
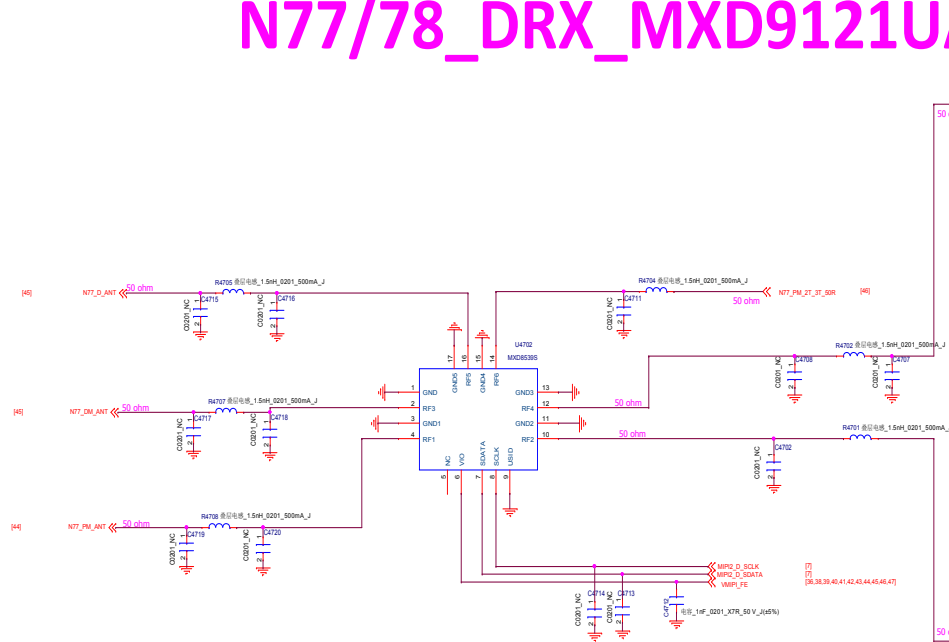
N77/78_MIMO_MXD9121UA



N77_78 PRX MIMO

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N77/78_DRX_MXD9121UA

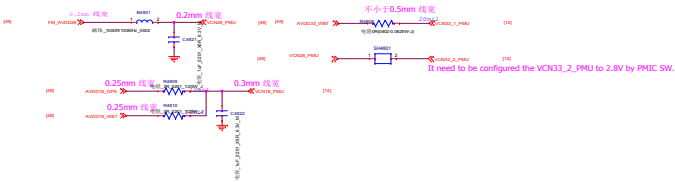
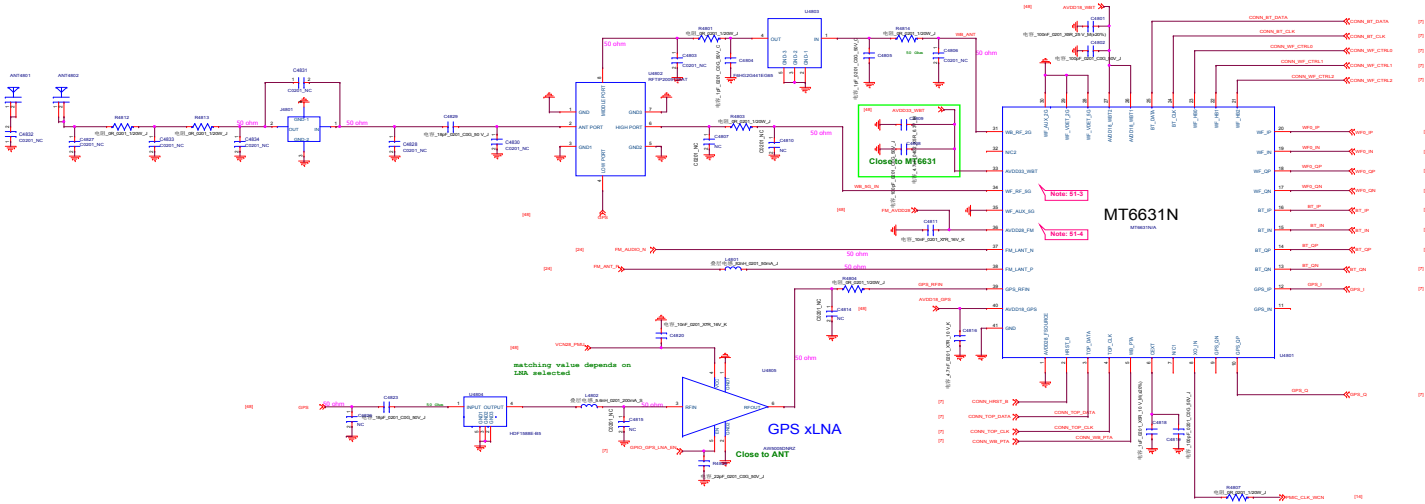


N77_78 DRX

N77_78 DRX MIMO

ANT7
WiFi 2.4G & 5G_GPS

MT6631



Schematic design notice of "51_CONNECTIVITY_CONSYS_MT6631"	
Note 51-1:	For R5015 size, please select 0402 size or larger one
Note 51-2:	Please refer to MT6765 Baseband design notice for VCN33_LDO selection guide
Note 51-3:	If WiFi 5G not support, connect pin 34(WF_RF_5G) to GND
Note 51-4:	Pin 36 (AVDD28_FM) must be connected to VCN28 even if FM not support

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MT6631		
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